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Radioactive decay and nuclear radiation

Atomic structure
Paper 1 · Combined and Separate
26 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State which part of an atom nuclear radiation is emitted from.**	The nucleus	☐
2	**Explain why unstable nuclei emit radiation.**	To become more stable.	☐
3	**What is radioactive decay?**	The random process in which an unstable nucleus emits radiation.	☐
4 ★	**Explain why radioactive decay is described as random.**	It is impossible to predict which nucleus will decay next, or when.	☐
5 ★	**Define the activity of a radioactive source.**	The rate at which its unstable nuclei decay.	☐
6 ★	**State the standard unit of activity.**	Becquerel (Bq)	☐
7	**What is count-rate?**	The number of decays recorded each second by a detector.	☐
8 ★	**Explain why the count-rate is always less than the activity.**	The detector picks up only some of the radiation emitted.	☐
9	**Name the four types of nuclear radiation.**	Alpha particles, beta particles, gamma rays and neutrons	☐
10	**Name three types of ionising nuclear radiation.**	Alpha, beta and gamma	☐
11 ★	**State what an alpha particle is made of.**	Two protons and two neutrons (a helium nucleus)	☐
12 ★	**State what a beta particle is.**	A high-speed electron from the nucleus	☐
13 ★	**State what a gamma ray is.**	Electromagnetic radiation from the nucleus	☐

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	QUESTION	ANSWER	✓
14	**A nucleus contains no electrons. Describe how it can emit a beta particle.**	A neutron turns into a proton and an electron; the electron is emitted.	☐
15 ★	**List alpha, beta and gamma radiation from most to least ionising.**	Alpha, beta, gamma	☐
16 ★	**List alpha, beta and gamma radiation from most to least penetrating.**	Gamma, beta, alpha	☐
17 ★	**State the range of alpha radiation in air.**	A few centimetres	☐
18	**State the range of beta radiation in air.**	About a metre	☐
19	**State the range of gamma radiation in air.**	A very long distance (it is hardly absorbed by air)	☐
20	**Name the type of radiation stopped by a sheet of paper.**	Alpha	☐
21 ★	**Name the two types of radiation stopped by a few millimetres of aluminium.**	Alpha and beta	☐
22 ★	**State what is needed to absorb most gamma radiation.**	Several centimetres of lead	☐
23	**Name the type of radiation stopped by skin.**	Alpha	☐
24	**Name a detector of nuclear radiation.**	A Geiger–Müller tube	☐
25	**State the relative charges of alpha, beta and gamma radiation.**	Alpha: +2 Beta: −1 Gamma: 0	☐
26	**What is ionisation?**	An atom gaining or losing electrons to become an ion.	☐

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